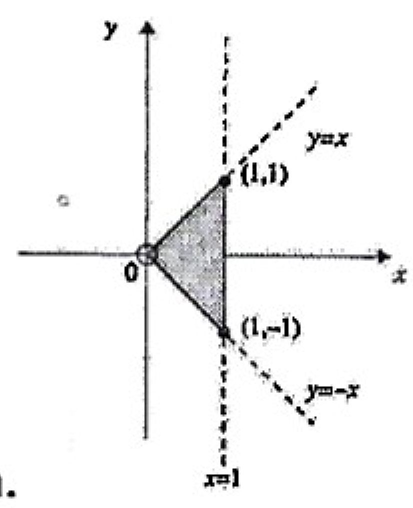
Complex Number 19

1. a. Find, in mod-arg form, the five fifth roots of i
   b. Find, in mod-arg form, the four fourth roots of - i
   c. Find, in the form $a + ib$, the four fourth roots of $-8 - 8\sqrt{3}i$
   d. Find, in mod-arg form, the five fifth roots of $16\sqrt{2} - 16\sqrt{2}i$
2. a. Find the five fifth roots of -1, writing the complex roots in mod-arg form
   b. If α is the complex root with least positive principal argument, show that α^3, α^7 and α^9 are the other three complex roots.
   c. Show that $\alpha^7 = -\alpha^2$ and that $\alpha^9 = -\alpha^4$
   d. Use the sum of the roots to show that $\alpha + \alpha^3 = 1 + \alpha^2 + \alpha^4$
3. a. Find roots of $z^7 - 1 = 0$.
   b. Hence show that: $\cos\frac{\pi}{7} - \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} = \frac{1}{2}$
   c. Write $z^7 - 1$ as a product of one linear and three quadratic factors, all with real coefficients
   d. If α is the complex seventh roots of unity with the least positive argument, show that $\alpha^2, \alpha^3, \alpha^4, \alpha^5$ and α^6 are the other five complex roots.
   e. By considering the relationships between the roots and the coefficients, show that the cubic equation $x^3 + x^2 - 2x - 1 = 0$ has roots $\alpha + \alpha^6, \alpha^2 + \alpha^5$ and $\alpha^3 + \alpha^4$.
4. If $z = 2 - i$. Find the real numbers p and q such that $pz + \frac{q}{z} = 1$.
5. z_1 and z_2 are complex numbers such that $|z_1| = 1$ and $|z_2| = 4$ and $\arg\frac{z_2}{z_1} = \frac{2\pi}{3}$
   a. Show $|\sqrt{z_1} - \sqrt{z_2}| = \sqrt{3}$.
   b. If $\arg\sqrt{z_1} = \alpha$ and $\left(0 < \alpha < \frac{\pi}{2}\right)$ find $\arg(\sqrt{z_1} - \sqrt{z_2})$ in terms of α.
6. a. If $z = \cos\theta + i\sin\theta$, prove
      i. $z^n + \frac{1}{z^n} = 2\cos n\theta$
      ii. $z^n - \frac{1}{z^n} = 2i\sin n\theta$
   b. By Letting $z = \cos\theta + i\sin\theta$ and expanding z^5 prove that
      i. $\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$
      ii. $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$
      iii. Using your workings for part I and ii find an expression for $\tan 5\theta$ in terms of $\tan\theta$.
   c. Prove that $\sin^5\theta = \frac{1}{16}[\sin 5\theta - 5\sin 3\theta + 10\sin\theta]$
7. a. i. Solve the equation $z^9 - 1 = 0$ over the complex field leaving your solutions in mod/arg form.
      ii. Find the area of the nonagon formed by joining the roots of $z^9 - 1 = 0$ on the complex plane. Answer correct to 2 decimal places.
   b. Let w be the non real root of $z^9 - 1 = 0$ with the smallest positive argument.
      i. Show that $w^2, w^3, w^4, w^5, w^6, w^7, w^8$ are the other non real roots of $z^9 - 1 = 0$.
      ii. Show that $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$.
      iii. Show that $(w + w^8)(w^2 + w^7)(w^4 + w^5) = -1$
      iv. Hence show that $\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{4\pi}{9} = \frac{1}{8}$.
   c. i. Resolve $z^5 - 1$ into real linear and quadratic factors. (Leave your quadratic factors in terms of $\cos\theta$)
      ii. Hence show that $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$ and $\cos\frac{2\pi}{5}\cos\frac{4\pi}{5} = -\frac{1}{4}$.
      iii. Hence find the exact value of $\cos\frac{2\pi}{5}$.

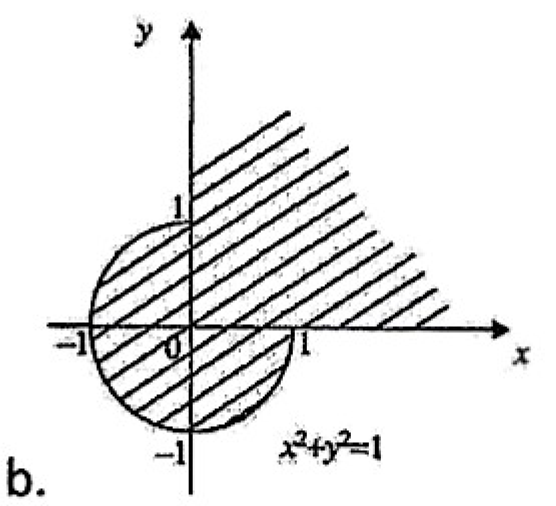
8. Indicate on an Argand diagram the region which contains the point P representing z when
   a. $|z| \leq |z-2|$ and $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$
   b. $|z| \leq 1$ or $0 \leq \arg z \leq \frac{\pi}{2}$
9. Express $(3+2i)(5+4i)$ and $(3-2i)(5-4i)$ in the form $a+ib$. Hence find the prime factors of 7^2+22^2.
10. Complex numbers $z_1 = \frac{a}{1+i}$ and $z_2 = \frac{b}{1+2i}$, where a and b are real, are such that $z_1 + z_2 = 1$. Find a and b.
11. $1+i$ is a root of the equation $x^2+(a+2i)x+(5+ib)=0$, where a and b are real. Find the values of a and b.
12. $1-2i$ is one root of the equation $x^2+(1+i)x+k=0$. Find the other root and the value of k.
13. a and b are real numbers such that the sum of the squares of the roots of the equation $x^2+(a+ib)x+3i=0$ is 8. Find all possible pairs of values a, b.
14. Solve $x^2-4x+(1-4i)=0$.
15. a. Find the roots of $z^5+1=0$ and show their position on a unit circle in an Argand diagram.
    b. Since the sum of these roots add to zero show $\cos\frac{\pi}{5}+\cos\frac{3\pi}{5}=\frac{1}{2}$.
16. The locus of a point $P(x,y)$, which moves in the complex plane is represented by the equation $|z-3i|=2$.
    a. Draw this locus, give a geometrical description and hence write down its Cartesian equation
    b. Find the modulus of z when P is in the position of minimum argument.
17. Show that, if the four points representing the complex numbers z_1, z_2, z_3, z_4 are concyclic, the fraction $\frac{(z_1-z_2)(z_3-z_4)}{(z_3-z_2)(z_1-z_4)}$ must be real.

Answer

1. a. $cis\left(-\frac{7\pi}{10}\right), cis\left(-\frac{3\pi}{10}\right), cis\frac{\pi}{10}, cis\frac{\pi}{2}=i, cis\frac{9\pi}{10}$;
   b. $cis\left(-\frac{\pi}{8}\right), cis\left(\frac{3\pi}{8}\right), cis\left(\frac{7\pi}{8}\right), cis\left(\frac{11\pi}{8}\right)$;
   c. $1+\sqrt{3}i, -1-\sqrt{3}i, \sqrt{3}-i, -\sqrt{3}+i$;
   d. $2cis\left(-\frac{17\pi}{20}\right), 2cis\left(-\frac{9\pi}{20}\right), 2cis\left(-\frac{\pi}{20}\right), 2cis\frac{7\pi}{20}, 2\,cis\frac{3\pi}{4}$
2. a. $-1, cis\frac{\pi}{5}, cis\left(-\frac{\pi}{5}\right), cis\frac{3\pi}{5}, cis\left(-\frac{3\pi}{5}\right)$
3. a. 1; $cis\left(\pm\frac{2\pi}{7}\right), cis\left(\pm\frac{4\pi}{7}\right), cis(\pm\frac{6\pi}{7})$; c. $(z-1)\left(z^2-2\cos\frac{2\pi}{7}z+1\right)\left(z^2-2\cos\frac{4\pi}{7}z+1\right)\left(z^2-2\cos\frac{6\pi}{7}z+1\right)$
4. $p=\frac{1}{4}, q=\frac{5}{4}$
5. b. $\arg(\sqrt{z_1}-\sqrt{z_2})=\alpha-\frac{\pi}{2}$
7. a. i. 1, $cis\left(\frac{2\pi}{9}\right)$, $cis\left(\frac{4\pi}{9}\right)$, $cis\left(\frac{2\pi}{3}\right)$, $cis\left(\frac{8\pi}{9}\right)$, $cis\left(\frac{10\pi}{9}\right)$, $cis\left(\frac{4\pi}{3}\right)$, $cis\left(\frac{14\pi}{9}\right)$, $cis\left(\frac{16\pi}{9}\right)$, ii. Area = 2.89 square units (2 d.p.); c. i. $z^5-1=(z-1)\left(z^2-2z\cos\frac{2\pi}{5}+1\right)\left(z^2-2z\cos\frac{4\pi}{5}+1\right)$, iii. $\frac{-1+\sqrt{5}}{4}$
8.



a.



b.

9. $7+22i, 7-22i$; $7^2+22^2=(3^2+2^2)(5^2+4^2)$
10. $a=4, b=-5$
11. $a=-3, b=-1$
12. $-2+i; k=5i$
13. $a=3, b=1; a=-3, b=-1$
14. $x=4+i, x=-i$
15. a. -1, $cis\left(\frac{\pi}{5}\right)$, $cis\left(-\frac{\pi}{5}\right)$, $cis\left(\frac{3\pi}{5}\right)$, $cis\left(-\frac{3\pi}{5}\right)$
16. a. radius= 2, centre = $(0,3)$, $x^2+(y-3)^2=4$; b. $|z|=\sqrt{5}$